

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical
PRACTICAL - 11

AIM : Reshaping data using pivot_longer()/pivot_wider() (R).

Input :

```
# R Script: Reshaping Data with pivot_longer() and pivot_wider()
# Dataset: loan_approval_dataset.csv
library(dplyr)
library(tidyr)
# 1. SETUP: Import Data
df <- read.csv("loan_approval_dataset.csv", na.strings = c("", "NA"))

# Add a unique ID for tracking during reshaping
df <- df %>% mutate(Applicant_ID = row_number())
cat("\n--- 1. Original Wide Data (First 6 Rows) ---\n")
print(head(df))

# 2. PIVOT_LONGER (Wide to Long)
# Automatically detect numeric columns for reshaping
num_cols <- names(df)[sapply(df, is.numeric)]
# Remove the ID column from pivoting
num_cols <- setdiff(num_cols, "Applicant_ID")
long_df <- df %>%
  pivot_longer(
    cols = all_of(num_cols),
    names_to = "Metric",
    values_to = "Value"
  )
cat("\n--- 2. Long Format (pivot_longer) ---\n")
print(head(long_df, 10))

# 3. PIVOT_WIDER (Long to Wide)
wide_df <- long_df %>%
```

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```
pivot_wider(  
  names_from = Metric,  
  values_from = Value  
)  
cat("\n--- 3. Wide Format (pivot_wider - Back to Original) ---\n")  
print(head(wide_df))
```

4. ADVANCED EXAMPLE – Category-wise Loan Report

Choose a category column safely (Gender / Education / Property_Area)

```
cat_cols <- names(df)[sapply(df, is.character)]
```

Pick the first category column automatically

```
category_col <- cat_cols[1]
```

```
cat("\n--- Category used for reporting ---\n")
```

```
print(category_col)
```

```
category_pivot <- df %>%
```

```
  select(Applicant_ID, all_of(category_col), all_of(num_cols[1])) %>%
```

```
  pivot_wider(  
    names_from = all_of(category_col),  
    values_from = all_of(num_cols[1])  
  )
```

```
cat("\n--- 4. Category-based Pivot Table ---\n")
```

```
print(head(category_pivot))
```

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Output :

The screenshot displays the RStudio interface with the following content:

Source

```
R - R4.5.2 - C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS/SPSS/R/Practical 11/

--- 1. Original Wide Data (First 6 Rows) ---
loan_id no_of_dependents education self_employed income_annum loan_amount loan_term cibil_score residential_assets_value
1 1 2 Graduate No 9600000 29900000 12 778 2400000
2 2 0 Not Graduate Yes 4100000 12200000 8 417 2700000
3 3 3 Graduate No 9100000 29700000 20 506 7100000
4 4 3 Graduate No 8200000 30700000 8 467 18200000
5 5 0 Not Graduate Yes 9800000 24200000 20 382 12400000
6 6 0 Graduate Yes 4800000 13500000 10 319 6800000

commercial_assets_value luxury_assets_value bank_asset_value loan_status Applicant_ID
1 17600000 22700000 8000000 Approved 1
2 2200000 8800000 3300000 Rejected 2
3 4500000 33300000 12800000 Rejected 3
4 3300000 23300000 7900000 Rejected 4
5 8200000 29400000 5000000 Rejected 5
6 8300000 13700000 5100000 Rejected 6

--- 2. Long Format (pivot_longer) ---
# A tibble: 10 x 6
  education self_employed loan_status Applicant_ID Metric Value
  <chr> <chr> <chr> <int> <chr> <int>
1 "Graduate" "No" "Approved" 1 loan_id 1
2 "Graduate" "No" "Approved" 1 no_of_dependents 2
3 "Graduate" "No" "Approved" 1 income_annum 9600000
4 "Graduate" "No" "Approved" 1 loan_amount 29900000
5 "Graduate" "No" "Approved" 1 loan_term 12
6 "Graduate" "No" "Approved" 1 cibil_score 778
7 "Graduate" "No" "Approved" 1 residential_assets_value 2400000
8 "Graduate" "No" "Approved" 1 commercial_assets_value 17600000
9 "Graduate" "No" "Approved" 1 luxury_assets_value 22700000
10 "Graduate" "No" "Approved" 1 bank_asset_value 8000000

--- 3. Wide Format (pivot_wider - Back to Original) ---
# A tibble: 6 x 14
  education self_employed loan_status Applicant_ID loan_id no_of_dependents income_annum loan_amount loan_term cibil_score
  <chr> <chr> <chr> <int> <int> <int> <int> <int> <int> <int>
1 "Graduate" "No" "Approved" 1 1 2 9600000 29900000 12 778
2 "Not Graduate" "Yes" "Rejected" 2 2 0 4100000 12200000 8 417
3 "Graduate" "No" "Rejected" 3 3 3 9100000 29700000 20 506
4 "Graduate" "No" "Rejected" 4 4 3 8200000 30700000 8 467

--- 3. Wide Format (pivot_wider - Back to Original) ---
# A tibble: 6 x 14
  education self_employed loan_status Applicant_ID loan_id no_of_dependents income_annum loan_amount loan_term cibil_score
  <chr> <chr> <chr> <int> <int> <int> <int> <int> <int> <int>
1 "Graduate" "No" "Approved" 1 1 2 9600000 29900000 12 778
2 "Not Graduate" "Yes" "Rejected" 2 2 0 4100000 12200000 8 417
3 "Graduate" "No" "Rejected" 3 3 3 9100000 29700000 20 506
4 "Graduate" "No" "Rejected" 4 4 3 8200000 30700000 8 467
5 "Not Graduate" "Yes" "Rejected" 5 5 5 9800000 24200000 20 382
6 "Graduate" "Yes" "Rejected" 6 6 0 4800000 13500000 10 319

# 4 more variables: residential_assets_value <int>, commercial_assets_value <int>, luxury_assets_value <int>, bank_asset_value <int>

--- Category used for reporting ---
[1] "education"

--- 4. Category-based Pivot Table ---
# A tibble: 6 x 3
  Applicant_ID `Graduate` `Not Graduate`
  <int> <int> <int>
1 1 1 NA
2 2 NA 2
3 3 3 NA
4 4 4 NA
5 5 NA 5
6 6 NA NA
```

Environment

Object	Variables
category_pi...	4269 obs. of 3 variables
df	4269 obs. of 14 variables
loan_approv...	4269 obs. of 13 variables
long_df	4269 obs. of 6 variables
wide_df	4269 obs. of 14 variables

Values

cat_cols	chr [1:3]	"education" "self_emplo...
category_col	chr [1:10]	"education"
num_cols	chr [1:10]	"loan_id" "no_of_depen..."